

* Rank 0 tensor. * Rank 1 tensor. * Rank 2 tensor (s).

$$\begin{bmatrix} 0 & G_{xy} & 0 \\ G_{xy} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$G_{xy} = \mathbb{G}_{xy}^{(1)}$$

* Rank 2 tensor (a). * Rank 3 tensor (ss). * Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ G_{yzx} & 0 & 0 \\ 0 & -G_{yzx} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$G_{yzx} = 3\mathbb{Q}_u^{(1)}$$

* Rank 3 tensor (a).

$$\begin{bmatrix} G_{yzx} & 0 & 0 \\ 0 & G_{yzx} & 0 \\ 0 & 0 & G_{xyz} \end{bmatrix}$$

$$G_{yzx} = \mathbb{Q}_0^{(1)} - \mathbb{Q}_u^{(2)}$$

$$G_{xyz} = \mathbb{Q}_0^{(1)} + 2\mathbb{Q}_u^{(2)}$$

* Rank 4 tensor (sss).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & G_{xxxy} \\ 0 & 0 & 0 & 0 & 0 & G_{xxxy} \\ 0 & 0 & 0 & 0 & 0 & G_{zzxy} \\ 0 & 0 & 0 & 0 & G_{yzzx} & 0 \\ 0 & 0 & 0 & G_{yzzx} & 0 & 0 \\ G_{xxxy} & G_{xxxy} & G_{zzxy} & 0 & 0 & 0 \end{bmatrix}$$

$$G_{xxxy} = \mathbb{G}_{xy}^{(1)} + 2\mathbb{G}_{xy}^{(2)} - \mathbb{G}_z^{\beta(1)}$$

$$G_{zzxy} = \mathbb{G}_{xy}^{(1)} + 2\mathbb{G}_z^{\beta(1)}$$

$$G_{yzzx} = \mathbb{G}_{xy}^{(2)} + 2\mathbb{G}_z^{\beta(1)}$$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & G_{xxxy} \\ 0 & 0 & 0 & 0 & 0 & G_{xxxy} \\ 0 & 0 & 0 & 0 & 0 & G_{zzxy} \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ -G_{xxxy} & -G_{xxxy} & -G_{zzxy} & 0 & 0 & 0 \end{bmatrix}$$

$$G_{xxxy} = \mathbb{G}_{xy}^{(3)} + 2\mathbb{Q}_z^{\beta(1)}$$

$$G_{zzxy} = \mathbb{G}_{xy}^{(3)} - 4\mathbb{Q}_z^{\beta(1)}$$

* Rank 4 tensor (aas).

$$\begin{bmatrix} 0 & G_{yzzx} & 0 \\ G_{yzzx} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$G_{yzzx} = 2\mathbb{G}_{xy}^{(4)}$$

* Rank 4 tensor (aaa). * Rank 4 tensor (sa).

$$\begin{bmatrix} 0 & 0 & G_{xxy} \\ 0 & 0 & -G_{xxy} \\ 0 & 0 & 0 \\ 0 & G_{yzzx} & 0 \\ -G_{yzzx} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$G_{xxy} = -2\mathbb{G}_{xy}^{(5)} + \mathbb{Q}_z^{\beta(2)}$$

$$G_{yzzx} = -\mathbb{G}_{xy}^{(5)} - \mathbb{Q}_z^{\beta(2)}$$

* Rank 4 tensor (as).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & G_{yzzx} & 0 \\ 0 & 0 & 0 & -G_{yzzx} & 0 & 0 \\ G_{xyxx} & -G_{xyxx} & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$G_{yzzx} = \mathbb{G}_{xy}^{(6)} + \mathbb{Q}_z^{\beta(3)}$$

$$G_{xyxx} = -2\mathbb{G}_{xy}^{(6)} + \mathbb{Q}_z^{\beta(3)}$$

* Rank 4 tensor (s).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & G_{xxy} & 0 & 0 & G_{xxyx} \\ 0 & 0 & 0 & 0 & 0 & G_{xxyx} & 0 & 0 & G_{xxy} \\ 0 & 0 & 0 & 0 & 0 & G_{zzxy} & 0 & 0 & G_{zzxy} \\ 0 & 0 & 0 & 0 & G_{yzzx} & 0 & 0 & G_{yzxz} & 0 \\ 0 & 0 & 0 & G_{yzxz} & 0 & 0 & G_{yzzx} & 0 & 0 \\ G_{xyxx} & G_{xyxx} & G_{xyzz} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$G_{xxy} = \mathbb{G}_{xy}^{(1)} + 2\mathbb{G}_{xy}^{(2)} + \mathbb{G}_{xy}^{(3)} - 2\mathbb{G}_{xy}^{(5)} - \mathbb{G}_z^{\beta(1)} + 2\mathbb{Q}_z^{\beta(1)} + \mathbb{Q}_z^{\beta(2)}$$

$$G_{xxyx} = \mathbb{G}_{xy}^{(1)} + 2\mathbb{G}_{xy}^{(2)} + \mathbb{G}_{xy}^{(3)} + 2\mathbb{G}_{xy}^{(5)} - \mathbb{G}_z^{\beta(1)} + 2\mathbb{Q}_z^{\beta(1)} - \mathbb{Q}_z^{\beta(2)}$$

$$G_{zzxy} = \mathbb{G}_{xy}^{(1)} + \mathbb{G}_{xy}^{(3)} + 2\mathbb{G}_z^{\beta(1)} - 4\mathbb{Q}_z^{\beta(1)}$$

$$G_{yzzx} = \mathbb{G}_{xy}^{(2)} - \mathbb{G}_{xy}^{(5)} + 2\mathbb{G}_z^{\beta(1)} - \mathbb{Q}_z^{\beta(2)}$$

$$G_{yzxz} = \mathbb{G}_{xy}^{(2)} + \mathbb{G}_{xy}^{(5)} + 2\mathbb{G}_z^{\beta(1)} + \mathbb{Q}_z^{\beta(2)}$$

$$G_{xyxx} = \mathbb{G}_{xy}^{(1)} + 2\mathbb{G}_{xy}^{(2)} - \mathbb{G}_{xy}^{(3)} - \mathbb{G}_z^{\beta(1)} - 2\mathbb{Q}_z^{\beta(1)}$$

$$G_{xyzz} = \mathbb{G}_{xy}^{(1)} - \mathbb{G}_{xy}^{(3)} + 2\mathbb{G}_z^{\beta(1)} + 4\mathbb{Q}_z^{\beta(1)}$$

* Rank 4 tensor (a).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & G_{yzzx} & 0 & 0 & G_{yzxz} & 0 \\ 0 & 0 & 0 & -G_{yzxz} & 0 & 0 & -G_{yzzx} & 0 & 0 \\ G_{xyxx} & -G_{xyxx} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$G_{yzzx} = 2\mathbb{G}_{xy}^{(4)} + \mathbb{G}_{xy}^{(6)} + \mathbb{Q}_z^{\beta(3)}$$

$$G_{yzxz} = -2\mathbb{G}_{xy}^{(4)} + \mathbb{G}_{xy}^{(6)} + \mathbb{Q}_z^{\beta(3)}$$

$$G_{xyxx} = -2\mathbb{G}_{xy}^{(6)} + \mathbb{Q}_z^{\beta(3)}$$

* Rank 4 tensor (t).

$$\begin{bmatrix} 0 & G_{xxxy} & 0 \\ G_{xxxy} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & G_{zzxy} & 0 \\ G_{xxyx} & 0 & 0 \\ G_{yzzx} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & G_{xyyy} & 0 \\ 0 & 0 & -\frac{G_{xxxy}}{2} - \frac{G_{xxyx}}{2} + G_{xyyy} + G_{zzxy} \end{bmatrix}$$

$$G_{xxxy} = \mathbb{G}_{xy}^{(1)} + 2\mathbb{G}_{xy}^{(2)} + \mathbb{G}_{xy}^{(3)} - 2\mathbb{G}_{xy}^{(5)} - \mathbb{G}_z^{\beta(1)}$$

$$G_{zzxy} = \mathbb{G}_{xy}^{(1)} + \mathbb{G}_{xy}^{(3)} + 2\mathbb{G}_z^{\beta(1)}$$

$$G_{xxyx} = \mathbb{G}_{xy}^{(1)} + 2\mathbb{G}_{xy}^{(2)} + \mathbb{G}_{xy}^{(3)} + 2\mathbb{G}_{xy}^{(5)} - \mathbb{G}_z^{\beta(1)}$$

$$G_{yzzx} = \mathbb{G}_{xy}^{(2)} - \mathbb{G}_{xy}^{(5)} + 2\mathbb{G}_z^{\beta(1)}$$

$$G_{xyyy} = \mathbb{G}_{xy}^{(1)} + 2\mathbb{G}_{xy}^{(2)} - \mathbb{G}_{xy}^{(3)} - \mathbb{G}_z^{\beta(1)}$$